

Asymptotic: Request Preservation and Portable Evidence

An incremental audit for Wolfram Language 15 and Mathics3

Technical review for Vladimir Reshetnikov

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Evidence	Pinned source, focused native observations, Python fixtures
Scope boundary	No full Mathics execution or complete native suite

Abstract

This review follows a broad inspection of the package’s mathematical engines, Mathics adaptations, numerical evidence paths, documentation, and existing review registers. It reports three incremental findings rather than reissuing the substantial earlier backlog. The most consequential is a Mathics input transformation that rewrites **Element** inside caller-owned held data and can thereby change the assumptions selected by a delayed option program. Installing the actual pinned adapter into an otherwise native Wolfram package changed a public model’s leading coefficient from 1 to 2. The second finding is a contradictory numerical diagnostic: **LocalRoot** remains a source coordinate while its label says it becomes a power observable. Two native public-API examples confirm the mismatch without an incorrect numerical value. The third finding is a portability gap in the local documentation checker: a link outside the checkout is accepted when an unrelated host file exists.

The accompanying artifacts provide a narrow rewrite candidate, a guarded patch emitter, proposed Wolfram regressions, a checkout-contained link checker, and 23 passing Python test methods. Evidence is separated by execution environment: actual Wolfram 15 standalone runs, a Mathics-adapter isolation experiment in Wolfram, isolated candidate checks, and local Python fixtures are not treated as interchangeable. No full Mathics run, patched-package acceptance, general cross-runtime equivalence, or exhaustive mathematical correctness is claimed.

The archive contains this article’s editable source and PDF, candidate code, reproducible tests, and scoped evidence. It contains no checksum files, separate font files, or complete upstream distribution.

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1 Conclusions and priorities

The principal new correctness risk is at the request boundary, not in an asymptotic coefficient formula. The Mathics assumptions adapter attempts to protect a mathematical proposition from an interpreter simplification, but its syntactic traversal is broader than the documented operation. It can modify a program that computes that proposition. A correct downstream algebra engine then solves a different problem from the one the caller supplied.

The numerical diagnostic finding is materially smaller. Its result values are correct in the executed examples; the accompanying explanation is not. The documentation finding is a validation gap with a concrete false-positive fixture, not evidence that the current repository contains a broken published link. These distinctions determine the proposed repair order.

ID	Priority	Incremental finding	Strongest evidence
N01	High impact; narrow trigger	Held caller data are rewritten inside an assumptions option; a delayed program changes its selected parameter condition.	Actual adapter isolated in the pinned package on Wolfram 15: coefficient $1 \rightarrow 2$.
N02	Low; straightforward repair	A coordinate-description string contradicts the source-root and powered-observable fields.	Two pinned native public-API executions.
N03	Medium validation gap	A local-link check can pass because a file exists outside the checkout.	Exact upstream Python source, actual Pandoc, temporary-tree counterexamples.

Table 1: Finding severity describes impact and repair urgency, not an estimate of how often ordinary users encounter the trigger.

Repair N02 independently because its minimal change is a descriptive string, not a numerical algorithm. For N01, first preserve the demonstrated quoted-data cases, then specify which assumption syntax is safely transformed. Do not present a few passing examples as a solution to arbitrary higher-order Wolfram program rewriting. For N03, decide the publication root explicitly and enforce it at the existing parser’s destination-resolution boundary.

The development recommendations in this article are restricted to these findings and their acceptance obligations. Existing requests for broad transseries extensions, general resource policies, release infrastructure, and native/Mathics feature expansion are not new contributions of this review.

2 Scope, reproducibility, and novelty control

2.1 A fixed source snapshot

All repository conclusions refer to `8cee870994f506b501bae3ea6bd4a3a7edb895c1`. Its commit concerns retirement tombstones for earlier review packages. Source inspection used pinned GitHub reads, including the recursive repository inventory, rather than assuming that `main` stayed unchanged during the audit. The native public-API probes loaded the root standalone file from that same commit. The isolated Mathics experiment additionally loaded the pinned modular assumptions adapter. The artifact does not contain a complete local checkout.[1, 2, 3]

A repository commit identifies the intended source, but it does not by itself establish that every test was executed against every distribution. In particular, the successful native standalone observations do not establish modular-loader acceptance. Conversely, inspecting modular source does not prove that every generated distribution is

fresh. The evidence record retains these distinctions rather than collapsing them into a single compatibility claim.

2.2 What actually executed

The remote Wolfram evaluator eventually reported

15.0.0 for Linux x86 (64-bit) (May 6, 2026).

Several earlier calls failed with network errors. Later successful calls established the following observations: the source-position counterexample for N01; the pinned-package comparison before and after isolated installation of the real Mathics assumptions adapter; two pinned native numerical diagnostic examples; and four isolated checks of the proposed rewrite guard. Exact submitted expressions and returned text are saved in `evidence/native-observations.json`. That file is a manually saved transcript of tool-mediated observations, not a fabricated export from an upstream test runner.

Locally, 23 Python methods passed without skips: 19 documentation characterization/candidate-policy methods and four patch-emitter fixture methods. The documentation fixture is the complete upstream `validation/documentation_links.py`; its transcription was byte-verified internally against the Git blob identity returned by GitHub. The tests invoked actual Pandoc, not a simulated Markdown parser. Exact tool versions and the final test output are in the evidence directory.

Not executed: a complete Mathics kernel/package run, a full native suite, a modular-load acceptance run, a patched full package, or the candidate link validator against a complete repository checkout. The supplied `tests/ReviewRegressions.wlt` is a proposed suite, not a suite whose pass count is being claimed. The candidate rewrite was checked in isolation in Wolfram, not accepted on Mathics.

2.3 How duplicate findings were excluded

The exclusion map was the maintained consolidated review-status register and all five review-wave indexes. Selected original material, particularly the nearby discussions in reviews 36, 42, and 44, was read directly where it could change the novelty assessment. This was not an exhaustive rereading of every retained original article. The exact distinction between each new mechanism and nearby old topics is recorded in the crosswalk below and in the archive. [10, 11, 12, 13, 14, 15, 16]

N01 is not another report of ambient assumptions leaking into refinement, a computed option key being rejected, or native Mathics membership simplification being unsound. It concerns a different mechanism: a newly inspected protector changing the identity of quoted data used by the caller’s assumption-generating program. N02 is not another large-offset cancellation problem or a proposal to change the source-observable convention; it identifies a concrete string that misdescribes the implemented convention. N03 is not another PDF receipt, last-pass recorder, or build-ledger concurrency problem; it concerns the filesystem namespace used to certify a Markdown target.

The source audit also encountered attractive but already recorded subjects: modulus handling, absolute-value regularity, exact numerical seed restrictions, stationary-point branch proofs, graded flat tails, repeated affine recognition, truncation suffix searches, and native trigonometric error transport. These were excluded rather than renamed as new issues. Absence of an item from this article therefore does not imply it was overlooked or fixed.

3 Architectural context relevant to the findings

The package separates a stored finite expression from remainder, coordinate, branch, provenance, and operation metadata. Its public entry points cover ordinary power-log expansion and inversion, several specialized inverse families, explicit series calculus, and native backend delegation. The `GeneralizedSeries` container is

consequently more than a displayed polynomial. A caller may consume its expression, its remainder, a model association, or a numerical comparison independently. Those consumers impose different obligations on the request and result fields.[2, 7, 8]

For Wolfram execution, the main loader resolves normal implementation calls to system functionality. On Mathics it loads an additional context of adapters and several late transformations of package-owned definitions. The assumptions protector is one such late transformation. It wraps the package's held `catch` boundary so syntactically supplied assumptions can be protected before ordinary option evaluation. It does not globally replace the caller's `System`Element`. This locality is valuable, but it does not ensure that a rewrite within the boundary respects the caller's data.[2, 3]

A useful audit decomposition is

request syntax $\longrightarrow$ interpreted hypotheses $\longrightarrow$ mathematical result $\longrightarrow$ reported evidence.

N01 affects the first arrow. N02 affects the last. N03 concerns the analogous interpretation of documentation paths: a repository-relative link is checked in a host-filesystem namespace that is too broad for a checkout-portability claim. None of these requires disproving the Lagrange coefficient formula or an exact interval enclosure to matter to users.

This audit did not establish a new public mathematical wrong-answer defect in the inspected coefficient recurrences or remainder formulas. That is a bounded negative result, not a proof of correctness for those engines. The source coverage appendix identifies complete module reads, partial reads, and unexecuted paths; it is not a clean-bill-of-health table.

4 N01: The Mathics assumptions protector changes quoted caller data

4.1 The intended protection and the actual traversal

The adapter's source explains its purpose with a Mathics membership problem: `Element[Sin[a], Reals]` must not simply be replaced by `Element[a, Reals]`. Its held entry path preserves an inline membership condition before native option evaluation reaches the problematic rule. The implementation searches for `Assumptions` rules, finds occurrences of `System`Element`, and replaces occurrences under those rules' values with `AsymptoticAnalysis`Mathics`Element`. [3]

The decisive operation is structurally:

```
heads = Position[held, System`Element,
  {0, Infinity}, Heads -> True];
positions = Select[heads, (* inside an Assumptions rule value *)];
ReplacePart[held,
  (# -> AsymptoticAnalysis`Mathics`Element) & /@ positions]
```

There are two different omissions. First, `Heads -> True` includes heads in the search; it does not restrict matches to heads. A bare symbol used as an argument or as data is also found. A head position ends in part index zero, whereas an ordinary argument position need not. The implementation does not make this distinction. Second, the traversal does not stop at nested `HoldComplete` or similar quotation constructs. It therefore rewrites even an actual membership *head* when that head is deliberately stored as unevaluated data. Wolfram's documented position and holding semantics are consistent with the executed mechanical witness.[17, 18]

These are not merely potential extra evaluation counts. The protector can leave an option program evaluated the same number of times while changing which result that program produces.

4.2 A minimal semantic counterexample

Consider the following caller state and held option:

```
saved = HoldComplete[System`Element];
h = HoldComplete[
  Assumptions :> If[
    HoldComplete[System`Element] === saved,
    a == 1,
    a == 2]];
```

The original strict equality is true: its two operands are identical held symbols. The occurrence on the left is syntactically inside an assumptions value, while the value stored in `saved` was created outside the request. The protector modifies the former but not the latter.

Proposition 1. *The inspected transformation is not semantics-preserving for delayed assumption programs that inspect held occurrences of `System`Element`.*

Proof. Let S denote the held system symbol and let S' denote the corresponding held Mathics-context symbol. They are distinct Wolfram expressions. The original program chooses $a = 1$ because S is identical to its saved value. The transformed program compares S' with the unchanged saved S , so it chooses $a = 2$. Both branches are valid parameter-only mathematical assumptions; the transformation changes the chosen one. $\square$

The native primitive experiment returned

```
{{{1, 2, 1, 1, 1}}, a == 1, a == 2}
```

for the matching position, original option value, and rewritten option value. The position ends in 1, not 0. This directly exposes the difference between including heads and selecting only heads.

The witness does not claim that merely holding an expression makes it immune to an explicit structural transformation. The defect is the mismatch between the transformation's intended scope—protecting active mathematical membership—and its actual scope, which includes caller-owned quoted data.

4.3 A public model changes after installation of the actual adapter

The issue is connected to a public entry point rather than stopping at an invented abstract tree. `PowerLogModel` has a held public entry and enters `catch` before `powerLogModelEntry` evaluates its options. The assumptions protector therefore sees the delayed program in the preceding example.^[2]

In a native Wolfram 15 kernel, the pinned standalone was first loaded and this call was evaluated:

```
PowerLogModel[a x, {x, 0},
  Assumptions :> If[
    HoldComplete[System`Element] === saved,
    a == 1, a == 2]]["LeadingCoefficient"]
```

The leading coefficient was 1. The actual pinned `MathicsInputAssumptions.wl` was then loaded in the package's Private context, without installing the rest of the Mathics runtime adaptations. The same public call returned 2. The exact paired output was

```
{1, 2}
```

This is strong evidence of the adapter’s transformation defect. It is also a carefully limited experiment: it is an adapter isolation run *in Wolfram*, not a complete Mathics execution. The loader source establishes that Mathics installs the adapter, but actual end-to-end Mathics acceptance remains to be performed.

For the model $f(x) = ax$, the correct requested normalized leading coefficient under $a = 1$ is exactly 1. The value 2 is correct for the *altered* assumptions. Thus the result need not contradict its own retained assumptions to violate the caller’s request. This is not evidence that the package proved a false theorem from the hypotheses it actually retained, nor an allegation that its interval arithmetic is unsound.

4.4 Why a head-only repair is necessary but insufficient

Adding

```
Length[position] > 0 && Last[position] === 0
```

excludes the bare-symbol witness. It does not protect this different datum:

```
saved = HoldComplete[System`Element[Sin[z], Reals]];
Assumptions :> If[
  HoldComplete[System`Element[Sin[z], Reals]] === saved,
  a == 1, a == 2]
```

Here the occurrence really is a head, but it is inside a nested held expression. The same strict-identity argument applies if the transformation descends into that data. This second original-adapter public sequence was not executed in the audit; its role is a source-derived counterexample and a required regression. The isolated candidate was tested to leave its held syntax unchanged.

The outer `HoldComplete` used to transport the request is different from a nested hold supplied by the caller. A correct barrier policy must traverse the former and preserve the latter. Simply skipping all held expressions would also disable the intended inline-assumptions protection.

4.5 Candidate repair and its limits

The supplied candidate narrows the replacement set to positions satisfying three conditions:

membership head $\wedge$ within an assumptions value $\wedge$ outside nested quoted-data barriers.

It preserves nested `HoldComplete`, `Hold`, `HoldForm`, `Defer`, `HoldPattern`, and `Verbatim` subtrees. The relevant part of the implementation is:

```
opaque = Position[held,
  _HoldComplete | _Hold | _HoldForm | _Defer |
  _HoldPattern | _Verbatim,
  {1, Infinity}, Heads -> False];

(* Keep only actual head positions, inside an option value,
   and outside every opaque subtree. *)
```

Four isolated observations in Wolfram returned `True`: an active inline membership head was still protected; bare held-symbol data were unchanged; a nested held membership call was unchanged; and a membership expression outside an assumptions option was unchanged. These checks establish the candidate’s behavior on those four trees, not a general transformation theorem or Mathics compatibility.

An arbitrary delayed option value is a Wolfram program, not necessarily a literal Boolean proposition. It may generate expressions, apply a symbol as a function, use a custom holding head, inspect syntax, or release a

quoted expression later. No finite list of common quotation heads establishes that a blanket source rewrite preserves every such program. In particular, preserving a nested hold does not protect a membership expression that the caller later releases and evaluates through native Mathics rules.

The immediate candidate should therefore be described as a repair for the shown data-capture cases. A stronger design needs a declared boundary: protect recognized declarative predicate syntax at the held entry, preserve arbitrary program syntax, and either safely normalize or explicitly refuse unsupported hypothesis-generation forms before their native interpretation loses information. Do not evaluate an arbitrary program twice to discover its shape. That would introduce a separate side-effect and reproducibility problem.

4.6 Performance implications of the proposed guard

The original mechanism scans for option rules and symbol occurrences and then checks every candidate occurrence against every option position. The proposed straightforward guard adds comparisons against quotation-barrier positions. If there are E membership-symbol occurrences, A option positions, B barriers, and maximum path length d , a direct prefix-comparison implementation can perform $O(E(A + B)d)$ element comparisons after its tree scans. This is a source-level bound for the proposed algorithm, not a measured Wolfram or Mathics slowdown.

Ordinary requests normally contain few assumptions options, but a large held payload can make B or E large. The narrow repair should not be justified by an unsupported claim of negligible overhead. Once the semantic boundary is settled, a single traversal carrying “inside an option value” and “inside opaque data” state can avoid repeated prefix searches for a restricted syntax grammar. Alternatively, selected position prefixes can be represented in a trie. Any such optimization must preserve the transport-wrapper distinction and caller-held payloads; a faster incorrect rewrite is not an improvement.

The acceptance workload should include large inert held trees as unchanged controls, not just increasingly large mathematical expressions. This is a specific performance obligation created by the new guard, not a restatement of the repository’s general resource-budget backlog.

4.7 Acceptance obligations

The two quoted-data examples must retain the original selected coefficient on actual Mathics. Ordinary inline membership protection must remain effective. The correction must not mutate caller saved values or package-global `System` definitions. Delayed option side effects and custom holding constructs should be tested according to the declared admission policy, rather than silently assumed safe.

Repeat the public examples with the native Wolfram path, the Mathics path, standalone loading, and modular loading. The purpose is not to demand that all interpreters support every mathematical function identically; it is to ensure that a supported request is not changed by an adapter before the mathematics starts. The included patch emitter and proposed tests make this acceptance work narrow and reviewable.

5 N02: The numerical result describes two different quantities as one

5.1 The implemented coordinate and observable

The common numerical evidence routine solves in a source coordinate $u > 0$:

$$x = \xi + \sigma u, \quad \sigma \in \{-1, 1\}.$$

At a finite source endpoint, ξ is that endpoint. For the numerical path at source infinity, the shift is zero and u is the positive source magnitude. This numerical coordinate must not be confused with a reciprocal small coordinate used internally by an asymptotic expansion.[4]

Let r be the requested **Power**. The implemented fields have the following meaning:

Field	$r = 1$	$r \neq 1$
LocalRoot	Recovered u	Recovered u , unchanged
LocalApproximation	Approximation to u	Approximation to $(\sigma u)^r$
ReferenceObservable	Recovered source x	Recovered signed displacement power $(\sigma u)^r$

The internal error calculation correctly compares corresponding local quantities. For $r \neq 1$, it compares **LocalApproximation** with $(\sigma u)^r$, not with **LocalRoot**. The returned label nevertheless says that both **LocalRoot** and **LocalApproximation** become values of u^{Power} . This is false for the first field and omits the source side for the second.^[4]

5.2 Executed native witnesses

Use the exactly invertible function $f(x) = x$. Its simplicity eliminates truncation and conditioning as explanations for a field disagreement. The following pinned native call was executed at 30-digit working precision:

```
s = AsymptoticInverse[x, {x, 0}, {y, 3}, "Power" -> 2];
r = InverseNumericalCheck[s, 2, WorkingPrecision -> 30];
```

It returned a local root of 2 and both a local approximation and a reference observable of 4. The old label says the local root is a value of u^2 ; it is not. The numerical quantities themselves are exactly the intended ones.

The signed example is even more discriminating:

```
s = AsymptoticInverse[x, {x, 0}, {y, 4},
  "Power" -> 3, Direction -> "FromBelow"];
r = InverseNumericalCheck[s, -2, WorkingPrecision -> 30];
```

The observed fields were

LocalRoot = 2, **LocalApproximation** = -8, **ReferenceObservable** = -8, **SourceSide** = -1.

The numerical error was zero at the reported precision. Neither 2 nor -8 is $u^3 = 8$ in the way asserted by the label. The output is nevertheless consistent with $x = -u$ and the requested x^3 observable.

Both are actual public-API observations from the pinned standalone on Wolfram 15. They are not outcomes from a Python emulation of the package. They also do not establish that every numerical check family uses this common routine; Gamma/Barnes and transformed/special families have separate dispatch paths. The finding is attributed to the routine that supplies this exact string. [2, 4, 9]

5.3 Impact and minimal correction

The risk is to a downstream consumer trusting the diagnostic description. For example, subtracting **LocalRoot** from **LocalApproximation** for the square case would produce 2, even though the correct observable error is zero. This is an error a consumer could make from the label, not an assertion that the package currently performs that subtraction.

Replace only the descriptive string with:

```
x = SourceOffset + SourceSide u; LocalRoot is u. LocalApproximation is u for
Power 1 and (SourceSide u)^Power otherwise.
```

The supplied emitter performs that single literal replacement for N02. It does not change roots, seed selection, remainder scales, branch checks, precision, legacy aliases, or observable conventions. This keeps the correction separate from larger API redesign discussions already present in the review corpus.

An optional additive field could expose the local reference observable that is already used to calculate the error. That would help code compare like quantities without reconstructing the side and power. Such a field should be introduced only with a clear contract and unchanged existing keys. It is not necessary for the minimal fix and is not included in the emitted patch.

Acceptance should preserve the numerical fields in both examples and verify the corrected description. A left-side even power and a translated source endpoint are useful additional controls. A regression that checks only the absolute root cannot detect this descriptive error; a regression that checks only the string cannot establish that the described convention is still the one implemented.

6 N03: Local documentation validity depends on unrelated host files

6.1 The namespace used by the validator

The maintained documentation checker parses Markdown with Pandoc's GFM parser, collecting ordinary links, images, raw HTML links, and fragments. Its local destination check then resolves a path relative to the source document and asks whether that destination exists. For Markdown and HTML fragments it reads the resolved target to check the anchor. The function receives a list of pages but no repository root.^[5, 6]

The critical code is:

```
url = urlsplit(href)
if url.scheme or url.netloc:
    continue

destination = ((path.parent / unquote(url.path)).resolve()
               if url.path else path)
if not destination.exists():
    errors.append(...)
```

Resolving `..` and symlinks is useful, but resolution is not containment. The implementation does not subsequently require that the resolved target belong to the checkout. An absolute filesystem path is also accepted by the path join when it exists. Python documents these path-resolution operations; the concrete behavior was separately exercised here.^[19]

6.2 An executed false portability acceptance

The fixture creates this temporary tree:

```
WORKSPACE/
  outside.md
  repo/
    docs/
      Guide.md
```

The source document contains

```
[outside](../../outside.md#outside)
```

and `outside.md` contains a matching heading. The exact upstream checker, using actual Pandoc, returned:

```
{
  "MaintainedMarkdownPages": 1,
  "LocalLinks": 1,
  "LocalFragments": 1,
  "BrokenLocalLinks": 0
}
```

Deleting only `WORKSPACE/outside.md`, without changing anything below `repo`, changed the outcome to a missing-destination assertion. A separate characterization also confirmed acceptance of an existing absolute filesystem target. The raw parent-path trace is saved in `evidence/link-characterization.json`.

Proposition 2. *The inspected validator’s success is not a property of the checkout contents alone.*

Proof. Keep the entire checkout fixed and vary only the outside target in the fixture. The checker succeeds when the target exists and fails when it is removed. Therefore its Boolean outcome depends on state outside the checkout. $\square$

The conclusion is deliberately narrower than “the repository’s documentation is broken.” No current committed link with this defect was established. The checker implements host-filesystem existence, which is a coherent but weaker property than portable repository-local validity. It cannot justify the stronger interpretation without another boundary check.

6.3 A concrete root-aware policy

The supplied candidate retains the upstream Markdown parser and introduces an explicit checkout root. It checks source pages and local destinations against that root, verifies actual component spelling, rejects symlink components, and then performs the existing fragment check. It accepts ordinary parent-relative links that remain inside the checkout, such as `../README.md##overview`; it does not ban all uses of `...`

The implementation deliberately chooses a conservative publication policy. All traversed symlink components, including in-root symlinks, are rejected. A publishing system with known symlink handling may choose a less restrictive policy, but that is not silently equated with ordinary file navigation. The candidate also rejects local `file:` URLs, Windows drive paths, and unconfigured root-relative paths. Web and other external URLs remain outside this local check.

A leading slash in a website URL can have a valid site-root meaning. It should not be interpreted accidentally as the operating system root. A future site publisher can map it through an explicit publication-root parameter. The candidate refuses it until that policy exists; refusal is preferable to acceptance because a similarly named file happens to exist on the validator’s machine.

Exact component spelling is checked by enumerating parent directory entries. That is intended to prevent a case-insensitive filesystem from hiding a capitalization mismatch. The tests here ran on Linux, so they establish the candidate’s explicit spelling behavior there, not an executed Windows comparison. The original checker would already reject many wrong-case paths on a case-sensitive filesystem. Case checking is part of the candidate’s portability policy, not a separately counted native Windows defect.

6.4 Limits and performance of the candidate

Containment alone does not prove that an in-root file is tracked by Git or included in a release. A locally generated or untracked file may still satisfy an existence check. Run publication acceptance on a stable clean checkout or add a separate, explicitly defined publication inventory. The candidate does not silently claim to supply that inventory.

Likewise, this is not a security sandbox against a process changing symlinks or files between validation and use. Its purpose is deterministic checkout-local documentation validation under a stable-tree assumption, not race-free filesystem capability confinement. Making stronger security claims would require a different implementation and operating-system contract.

Directory name sets, parsed Markdown, and anchors are cached within one check. This avoids rescanning a directory or reparsing a target for every repeated link. If there are L local links with at most d path components, the path walk performs $O(Ld)$ membership checks after the relevant directory entries are collected; actual I/O cost also depends on directory sizes and the files Pandoc parses. This describes the candidate implementation, not a claimed measured speedup over the current checker. The same single-check cache must not be reused to certify a changed filesystem later.

6.5 Executed tests and integration requirements

The 19 documentation methods distinguish characterization from candidate acceptance. Three characterize the upstream dependency on external files. The others check the new policy with parent escapes, encoded escapes, absolute paths, valid in-root parent links, self fragments, HTML fragments, missing anchors, case mismatches, images, file URLs, excluded external URLs, internal and external symlinks, outside source files, escaped spaces, and missing destinations. All passed in the recorded environment, with no skips.

The complete repository was not run through the candidate. Integration should first pass the known root from the maintained-document discovery function into the local-link checker, then preserve the existing Markdown/HTML anchor semantics. Any rejected real link must be classified: an actual checkout escape, a legitimate separately published asset requiring an explicit mapping, or an intentionally external URL. Do not replace that classification with an unrestricted host-filesystem fallback.

7 Cross-runtime assessment and a bounded development plan

7.1 What the evidence says about Wolfram 15 and Mathics3

The two numerical witnesses establish that the native standalone works for those simple exact inverse cases on the observed Wolfram 15 kernel. They do not establish compatibility for every package feature or for every Wolfram 15 maintenance version. The isolated adapter run establishes a syntactic request mutation in the actual Mathics-specific module; it does not establish the complete behavior of Mathics with every other adapter enabled.

The right conclusion is neither “Mathics is incompatible” nor “the package is compatible because the coefficient mathematics is shared.” Compatibility must include the interpretation of supported requests and the meaning of result diagnostics, as well as successful coefficient construction. N01 is a particularly useful regression because its forward model is only ax ; a failure cannot be excused as unsupported advanced special-function algebra.

The source audit observed many bounded Mathics adaptations rather than a single global emulation layer. This limits the scope of each intervention, but the late input protector demonstrates that locality of installation does not itself guarantee locality of semantic effect. Preserve that distinction when assessing fixes.[2, 3]

7.2 Targeted implementation sequence

First, correct and lock down the numerical description. Apply the N02 string-only change and retain the power-two and signed odd-power observations as paired value/description regressions. Do not use this small issue to justify renaming established fields or changing source-observable conventions.

Second, repair quoted-data preservation at the assumptions boundary. Use the bare-symbol and nested-held-call cases as mandatory controls. The candidate is intentionally narrow. Specify the protected predicate grammar and the behavior of unsupported generated or higher-order forms before expanding the

transformation. Add large quoted payloads as structural-work controls so correctness is not purchased with avoidable repeated traversals.

Third, define the documentation publication namespace. Integrate the root-aware path policy with the existing Pandoc parser. Keep external URLs, site-root mappings, symlink publication, and Git-trackedness distinct. A local filesystem path should not count as a repository-local proof simply because it exists during one check.

These changes can be reviewed separately. N02 is a descriptive correction; N01 changes evaluation-boundary behavior; N03 changes validation admission. They need different rollback and acceptance criteria. A broad refactor tying all three together would make the mathematical and engineering evidence harder to interpret without providing a compensating benefit.

7.3 Further opportunities arising specifically from these findings

For the assumptions protector, develop metamorphic tests based on quotation: adding or moving unrelated held data must not alter the interpreted active hypotheses. A useful property is that replacing a saved quoted datum by another syntactically identical quoted datum outside the request leaves the selected assumptions invariant. Counterexamples should be minimized to a held program, not hidden inside expensive asymptotic examples.

For numerical diagnostics, specify quantity identities alongside labels. The error comparison should be reconstructible from fields that denote the same observable, including the selected sign. This can be checked on exact linear models before introducing numerical conditioning or asymptotic truncation. The opportunity is a small diagnostic contract test, not a general redesign of all result families.

For documentation, use paired workspaces with identical checkout contents and different unrelated parent files. Under a checkout-contained policy, outcomes must agree. The existing N03 fixture already supplies the smallest such metamorphic test. Expanding it to a clean publication tree is more informative than repeatedly running the checker on the same developer filesystem.

No native speedup, general performance regression, or improved asymptotic coverage is claimed from these recommendations. Their value is reducing semantic and environmental ambiguity in the three demonstrated boundaries.

8 Artifact guide and reproduction

The archive is arranged so the strongest independently reproducible checks do not require a Wolfram license. From the archive root, run:

```
python -m unittest discover -s tests -p 'test_*.py' -v
python code/characterize_links.py
```

Python 3.10 or newer and Pandoc are required. The two symlink tests skip when symlink creation is unavailable; a run with skips is not identical to the recorded no-skip run. The tests create temporary fixture trees and do not alter a user's repository. The candidate link checker can be pointed at an existing checkout without writing it:

```
python code/portable_links.py /path/to/Asymptotic
```

That full-checkout command was not executed here.

For source edits, the emitter checks the audited Git revision and refuses modified tracked target files before producing a diff:

```
python code/emit_candidate_patch.py /path/to/Asymptotic \
--which diagnostics > candidate.diff
```

Use `-which assumptions` for the narrow rewrite candidate, or `-which all` for both. The explicit `-skip-commit-check` switch is for a consciously chosen alternate tree; it does not make that tree part of this audit. All selected anchors are validated before any diff is printed. The emitter does not apply its output. Its four tests use synthetic source fixtures, not an actual upstream patch-application run.

In a fresh Wolfram session, load the known local standalone and then use `TestReport` on `tests/ReviewRegressions.wlt`. Baseline failures of correction tests are expected until the relevant fix is present. For Mathics, `code/probe_loaded_kernel.wl` prints observations without assuming MUnit support. The file is a probe, not an aggregate test gate. The isolated candidate can be loaded from `code/candidate_assumption_protector.wl` and examined with `code/check_candidate_rewrite.wl`; it does not install itself into the package.

Before merging a canonical-source change, regenerate the standalone through the repository’s existing builder and repeat the focused acceptance checks on the intended runtimes. This instruction describes how to validate these specific candidates; it is not a claim that the generated standalone in this archive has been patched or that any repository write occurred.

A Source coverage and residual uncertainty

The following records the audit’s main source coverage. “Inspected” is not synonymous with “executed” or “verified.” Source ranges are original file ranges, not line numbers of JSON tool wrappers. Full module reads were used where the entire returned file was available; several larger modules were inspected only in relevant ranges.

Area	Inspected material	Boundary of the conclusion
Public entries and loader	<code>AsymptoticAnalysis.wl</code> : lines 1–150 and approximately 1000–end, including constructor assembly, accessors, residual/numerical dispatch, model entry, and loader.	Main-file middle sections were not read exhaustively. Focused native standalone only.
Mathics request handling	<code>CompleteMathicsInputAssumptions.wl</code> , <code>MathicsCalls.wl</code> , <code>MathicsLists.wl</code> ; compatibility, assumptions, simplification, and late branch wrappers.	Actual input-adaptor isolation; no complete Mathics execution.
Other Mathics adapters	Algebra, Taylor, calculus, core functions, refinement, special functions, certificate assignment, and numerical adapters.	Source inspection; previously recorded defects excluded.
Ordinary numerical evidence	<code>CompleteNumericalInverseChecks.wl</code> ; native dispatch in the main module.	Two exact public witnesses; other functions and targets not exhaustively tested.
Inverse certificates	<code>InverseCertificates.wl</code> , inspected in ranges through the adaptive certificate implementation.	No new interval arithmetic defect established; no full certificate suite.
Gamma and Barnes families	Gamma forward/inverse modules, Barnes inverse module, Gamma inverse checks, and relevant special adapters.	Source analysis, not broad numerical acceptance.
Special forward functions	Dirichlet and parameterized special-function modules; exact-model and finite-model distinctions in adjacent adapters.	No new coefficient/remainder defect established; no full native importer audit.
Series operations	Initial representation and arithmetic sections of <code>SeriesOperations.wl</code> ; complete composite-envelope module; early flat-sector operations.	Partial operation coverage, not universal closure proof.

Area	Inspected material	Boundary of the conclusion
Fourier and flat sectors	Initial Fourier coefficient/model sections and graded flat-tail implementation.	Existing findings excluded; no new full-family runtime result.
Documentation tooling	Complete <code>documentation_links.py</code> and <code>build_user_guide.py</code> ; maintenance and validation documentation.	Exact local-link fixture execution; not complete repository documentation acceptance.
Prior reviews	Consolidated register, five wave indexes, selected source bodies/README material for neighboring reviews.	Novelty screening, not exhaustive rereading of all original articles.

The source ranges do not support claims about untouched algorithms, every parameter domain, all resource limits, or all loader failure paths. Additional coverage could find further issues. In particular, the absence of a local Mathics installation prevents turning N01’s source reachability and native isolation experiment into a full Mathics runtime observation.

B Evidence populations and acceptance status

Population	Actually obtained	Not implied
Native public numerical probes	Pinned standalone, square and left-side cube, Wolfram 15.0.0.	Other families, modular loading, complete Wolfram suite.
Actual Mathics input adapter	Before/after public model coefficient $1 \rightarrow 2$ after isolated module installation in Wolfram.	A complete Mathics kernel run.
Candidate rewrite algorithm	Four isolated Boolean observations returned true in Wolfram.	General option-program semantics or patched Mathics acceptance.
Upstream link checker	Exact transcription, actual Pandoc, outside-file and absolute-path counterexamples.	A current committed broken link.
Candidate link policy	Temporary-tree tests for the stated containment/spelling/symlink/fragment policy.	A full checkout pass, tracked-file inventory, race-free security boundary.
Patch emitter	Four source-fixture methods passed.	An actual upstream patch application or native patched run.
Proposed Wolfram suite	Supplied as editable test source.	Any suite pass count.

The first draft of an emitter-fixture test mistakenly chose the word **none** as a string not containing **one**. That fixture was corrected to **absent**; the final recorded 23-method run passed. This was a test-authoring error, not an upstream defect. The scope record preserves the distinction rather than attributing the first local failure to the repository.

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